Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.41)

Metric

$$ds^2 = -H^2 \mathrm{d}u \otimes \mathrm{d}u + \left(-\frac{l^2 - z^2}{l^2 + z^2}\right) \mathrm{d}u \otimes \mathrm{d}v + \left(\frac{kl^2vz - kvz^3 - 2\,mvz^2}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}u \otimes \mathrm{d}z + \left(-\frac{kl^3v - klvz^2 - 2\,lmvz}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}u \otimes \mathrm{d}y + \left(-\frac{l^2 - z^2}{l^2 + z^2}\right) \mathrm{d}v \otimes \mathrm{d}u + \left(\frac{kl^2vz - kvz^3 - 2\,mvz^2}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}z \otimes \mathrm{d}u + \mathrm{d}z \otimes \mathrm{d}z + \left(-\frac{kl^3v - klvz^2 - 2\,lmvz}{l^4 + 2\,l^2z^2 + z^4}\right) \mathrm{d}y \otimes \mathrm{d}u + \left(-\frac{l^2 + z^2}{kl^2 - kz^2 - 2\,mz}\right) \mathrm{d}y \otimes \mathrm{d}y$$

Coordinates

$(x^\mu) = (u, v, z, y)$

Metric Functions

$$H = \frac{2\,kl^4v^2}{l^6 + 3\,l^4z^2 + 3\,l^2z^4 + z^6} - \frac{2\,kl^2v^2z^2}{l^6 + 3\,l^4z^2 + 3\,l^2z^4 + z^6} - \frac{4\,l^2mv^2z}{l^6 + 3\,l^4z^2 + 3\,l^2z^4 + z^6} - \frac{kv^2}{2\,(l^2 + z^2)}$$

Null Tetrad

Covariant components

$l = (1)\ du$

$$n = \left(\frac{1}{2}\,H^2\right)\ du + \left(\frac{l^2}{l^2 + z^2} - \frac{z^2}{l^2 + z^2}\right)\ dv + \left(-\frac{kl^2vz}{l^4 + 2\,l^2z^2 + z^4} + \frac{kvz^3}{l^4 + 2\,l^2z^2 + z^4} + \frac{2\,mvz^2}{l^4 + 2\,l^2z^2 + z^4}\right)\ dz + \left(\frac{kl^3v}{l^4 + 2\,l^2z^2 + z^4} - \frac{klvz^2}{l^4 + 2\,l^2z^2 + z^4} - \frac{2\,lmvz}{l^4 + 2\,l^2z^2 + z^4}\right)\ dy$$

$$m = \left(\sqrt{\frac{1}{2}}\right)\ dz + \left(\frac{i\,\sqrt{\frac{1}{2}}\sqrt{l^2 + z^2}}{\sqrt{-kl^2 + kz^2 + 2\,mz}}\right)\ dy$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{\frac{1}{2}}\sqrt{l^2+z^2}}{\sqrt{-kl^2+kz^2+2\,mz}}\right) dy$$

Contravariant components

$$l = \left(-\frac{l^2}{l^2-z^2} - \frac{z^2}{l^2-z^2}\right) dv$$

$$n = (-1)\, du + \left(\frac{l^2H^2}{2\,(l^2-z^2)} + \frac{z^2H^2}{2\,(l^2-z^2)}\right) dv$$

$$\begin{aligned} m = \Bigg(& \frac{i\sqrt{2}\sqrt{l^2+z^2}k^2l^5v}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}k^2l^3vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}k^2lvz^4}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} + \frac{\sqrt{2}kl^4vz}{2\,(l^6+l^4z^2-l^2z^4-z^6)} \\ & - \frac{2i\sqrt{2}\sqrt{l^2+z^2}kl^3mvz}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} + \frac{2i\sqrt{2}\sqrt{l^2+z^2}klmvz^3}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{\sqrt{2}kvz^5}{2\,(l^6+l^4z^2-l^2z^4-z^6)} - \frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} \\ & + \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6}\Bigg) dv \\ & + \left(\sqrt{\frac{1}{2}}\right) dz + \left(-\frac{i\sqrt{2}kl^2}{2\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{i\sqrt{2}kz^2}{2\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{i\sqrt{2}mz}{\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\right) dy \end{aligned}$$

$$\begin{aligned} \overline{m} = \Bigg(& -\frac{i\sqrt{2}\sqrt{l^2+z^2}k^2l^5v}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}k^2l^3vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}k^2lvz^4}{2\,(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} + \frac{\sqrt{2}kl^4vz}{2\,(l^6+l^4z^2-l^2z^4-z^6)} \\ & + \frac{2i\sqrt{2}\sqrt{l^2+z^2}kl^3mvz}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{2i\sqrt{2}\sqrt{l^2+z^2}klmvz^3}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{\sqrt{2}kvz^5}{2\,(l^6+l^4z^2-l^2z^4-z^6)} - \frac{\sqrt{2}l^2mvz^2}{l^6+l^4z^2-l^2z^4-z^6} \\ & - \frac{2i\sqrt{2}\sqrt{l^2+z^2}lm^2vz^2}{(l^6+l^4z^2-l^2z^4-z^6)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{\sqrt{2}mvz^4}{l^6+l^4z^2-l^2z^4-z^6}\Bigg) dv \\ & + \left(\sqrt{\frac{1}{2}}\right) dz + \left(\frac{i\sqrt{2}kl^2}{2\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} - \frac{i\sqrt{2}kz^2}{2\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} - \frac{i\sqrt{2}mz}{\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\right) dy \end{aligned}$$

Directional Derivatives

$$D = -\frac{l^2}{l^2 - z^2} - \frac{z^2}{l^2 - z^2}\,\partial_v$$

$$\Delta = -\partial_u + \frac{l^2 H^2}{2\,(l^2 - z^2)} + \frac{z^2 H^2}{2\,(l^2 - z^2)}\,\partial_v$$

$$\begin{aligned}
\delta = {} & \frac{i\,\sqrt{2}\sqrt{l^2+z^2}k^2 l^5 v}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} - \frac{i\,\sqrt{2}\sqrt{l^2+z^2}k^2 l^3 v z^2}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} + \frac{i\,\sqrt{2}\sqrt{l^2+z^2}k^2 l v z^4}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} \\
& + \frac{\sqrt{2}kl^4 vz}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)} - \frac{2i\,\sqrt{2}\sqrt{l^2+z^2}kl^3 mvz}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} + \frac{2i\,\sqrt{2}\sqrt{l^2+z^2}klmvz^3}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} \\
& - \frac{\sqrt{2}kvz^5}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)} - \frac{\sqrt{2}l^2 mvz^2}{l^6 + l^4 z^2 - l^2 z^4 - z^6} + \frac{2i\,\sqrt{2}\sqrt{l^2+z^2}lm^2 vz^2}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} - \frac{\sqrt{2}mvz^4}{l^6 + l^4 z^2 - l^2 z^4 - z^6}\,\partial_v \\
& + \sqrt{\frac{1}{2}}\,\partial_z - \frac{i\,\sqrt{2}kl^2}{2\,\sqrt{-kl^2 + kz^2 + 2\,mz}\sqrt{l^2+z^2}} + \frac{i\,\sqrt{2}kz^2}{2\,\sqrt{-kl^2 + kz^2 + 2\,mz}\sqrt{l^2+z^2}} + \frac{i\,\sqrt{2}mz}{\sqrt{-kl^2 + kz^2 + 2\,mz}\sqrt{l^2+z^2}}\,\partial_y
\end{aligned}$$

$$\begin{aligned}
\overline{\delta} = {} & -\frac{i\,\sqrt{2}\sqrt{l^2+z^2}k^2 l^5 v}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} + \frac{i\,\sqrt{2}\sqrt{l^2+z^2}k^2 l^3 v z^2}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} - \frac{i\,\sqrt{2}\sqrt{l^2+z^2}k^2 l v z^4}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} \\
& + \frac{\sqrt{2}kl^4 vz}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)} + \frac{2i\,\sqrt{2}\sqrt{l^2+z^2}kl^3 mvz}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} - \frac{2i\,\sqrt{2}\sqrt{l^2+z^2}klmvz^3}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} \\
& - \frac{\sqrt{2}kvz^5}{2\,(l^6 + l^4 z^2 - l^2 z^4 - z^6)} - \frac{\sqrt{2}l^2 mvz^2}{l^6 + l^4 z^2 - l^2 z^4 - z^6} - \frac{2i\,\sqrt{2}\sqrt{l^2+z^2}lm^2 vz^2}{(l^6 + l^4 z^2 - l^2 z^4 - z^6)\sqrt{-kl^2 + kz^2 + 2\,mz}} - \frac{\sqrt{2}mvz^4}{l^6 + l^4 z^2 - l^2 z^4 - z^6}\,\partial_v \\
& + \sqrt{\frac{1}{2}}\,\partial_z + \frac{i\,\sqrt{2}kl^2}{2\,\sqrt{-kl^2 + kz^2 + 2\,mz}\sqrt{l^2+z^2}} - \frac{i\,\sqrt{2}kz^2}{2\,\sqrt{-kl^2 + kz^2 + 2\,mz}\sqrt{l^2+z^2}} - \frac{i\,\sqrt{2}mz}{\sqrt{-kl^2 + kz^2 + 2\,mz}\sqrt{l^2+z^2}}\,\partial_y
\end{aligned}$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{i\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}kl^5vz}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} - \frac{2i\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}kl^3vz^3}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} + \frac{i\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}klvz^5}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}}$$
$$+ \frac{i\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}l^5mv}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}} + \frac{3i\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}lmvz^4}{l^{10}+3\,l^8z^2+2\,l^6z^4-2\,l^4z^6-3\,l^2z^8-z^{10}}$$

$$\lambda = 0$$

$$\nu = -\frac{i\sqrt{2}k^2l^5vH\frac{\partial}{\partial v}H}{2\,(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{i\sqrt{2}k^2l^3vz^2H\frac{\partial}{\partial v}H}{(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} - \frac{i\sqrt{2}k^2lvz^4H\frac{\partial}{\partial v}H}{2\,(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{i\sqrt{2}k^2l^5H^2}{4\,(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}$$
$$- \frac{i\sqrt{2}k^2l^3z^2H^2}{2\,(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{i\sqrt{2}k^2lz^4H^2}{4\,(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{2i\sqrt{2}kl^3mvzH\frac{\partial}{\partial v}H}{(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} - \frac{2i\sqrt{2}klmvz^3H\frac{\partial}{\partial v}H}{(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}$$
$$- \frac{i\sqrt{2}kl^3mzH^2}{(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{i\sqrt{2}klmz^3H^2}{(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{\sqrt{2}kl^2vzH\frac{\partial}{\partial v}H}{2\,(l^4-z^4)} - \frac{2i\sqrt{2}lm^2vz^2H\frac{\partial}{\partial v}H}{(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} - \frac{\sqrt{2}kvz^3H\frac{\partial}{\partial v}H}{2\,(l^4-z^4)}$$
$$- \frac{\sqrt{2}kl^2zH^2}{4\,(l^4-z^4)} + \frac{i\sqrt{2}lm^2z^2H^2}{(l^4-z^4)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{\sqrt{2}kz^3H^2}{4\,(l^4-z^4)} + \frac{\sqrt{2}l^4H\frac{\partial}{\partial z}H}{2\,(l^4-z^4)} - \frac{\sqrt{2}z^4H\frac{\partial}{\partial z}H}{2\,(l^4-z^4)} - \frac{\sqrt{2}mvz^2H\frac{\partial}{\partial v}H}{l^4-z^4} + \frac{\sqrt{2}l^2zH^2}{l^4-z^4} + \frac{\sqrt{2}mz^2H^2}{2\,(l^4-z^4)}$$

$$
\begin{aligned}
\alpha = {} & \frac{i\sqrt{l^2+z^2}k^3l^7}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& - \frac{3i\sqrt{l^2+z^2}k^3l^5z^2}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& + \frac{3i\sqrt{l^2+z^2}k^3l^3z^4}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& - \frac{i\sqrt{l^2+z^2}k^3lz^6}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& - \frac{k^2l^6z}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)} \\
& - \frac{3i\sqrt{l^2+z^2}k^2l^5mz}{2\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& + \frac{k^2l^4z^3}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)} \\
& + \frac{3i\sqrt{l^2+z^2}k^2l^3mz^3}{\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& + \frac{k^2l^2z^5}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)} \\
& - \frac{3i\sqrt{l^2+z^2}k^2lmz^5}{2\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& - \frac{k^2z^7}{4\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)} + \frac{kl^4mz^2}{\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7} \\
& + \frac{3i\sqrt{l^2+z^2}kl^3m^2z^2}{\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& - \frac{3i\sqrt{l^2+z^2}klm^2z^4}{\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& - \frac{kmz^6}{\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7} - \frac{l^6m}{2\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)} \\
& - \frac{3\,l^4mz^2}{2\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)} - \frac{l^2m^2z^3}{\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7} \\
& - \frac{2i\sqrt{l^2+z^2}lm^3z^3}{\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)\sqrt{-kl^2+kz^2+2mz}} \\
& - \frac{3\,l^2mz^4}{2\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)} - \frac{m^2z^5}{\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7} \\
& - \frac{mz^6}{2\left(\sqrt{2}kl^8-2\sqrt{2}kl^4z^4+\sqrt{2}kz^8-2\sqrt{2}l^6mz-2\sqrt{2}l^4mz^3+2\sqrt{2}l^2mz^5+2\sqrt{2}mz^7\right)}
\end{aligned}
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$$
\begin{aligned}
\beta = &-\frac{i\,k^3l^7}{4\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&+\frac{3i\,k^3l^5z^2}{4\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&-\frac{3i\,k^3l^3z^4}{4\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&+\frac{i\,k^3lz^6}{4\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&+\frac{3i\,k^2l^5mz}{2\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&-\frac{3i\,k^2l^3mz^3}{\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&+\frac{3i\,k^2lmz^5}{2\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&-\frac{k^2l^4z}{4\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)}-\frac{3i\,kl^3m^2z^2}{\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&+\frac{k^2l^2z^3}{2\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)}+\frac{3i\,klm^2z^4}{\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&-\frac{k^2z^5}{4\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)}+\frac{2\,kl^4z}{\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5}\\
&+\frac{kl^2mz^2}{\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5}-\frac{2\,kl^2z^3}{\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5}\\
&+\frac{2i\,lm^3z^3}{\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}-\frac{kmz^4}{\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5}\\
&+\frac{l^4m}{2\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)}-\frac{3\,l^2mz^2}{\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5}\\
&-\frac{m^2z^3}{\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5}+\frac{mz^4}{2\left(\sqrt{2}kl^6-\sqrt{2}kl^4z^2-\sqrt{2}kl^2z^4+\sqrt{2}kz^6-2\sqrt{2}l^4mz+2\sqrt{2}mz^5\right)}
\end{aligned}
$$

$$
\begin{aligned}
\tau = &-\frac{i\sqrt{2}k^2l^5}{4\left(l^4-z^4\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}+\frac{i\sqrt{2}k^2l^3z^2}{2\left(l^4-z^4\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}-\frac{i\sqrt{2}k^2lz^4}{4\left(l^4-z^4\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}+\frac{i\sqrt{2}kl^3mz}{\left(l^4-z^4\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}\\
&-\frac{i\sqrt{2}klmz^3}{\left(l^4-z^4\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}-\frac{\sqrt{2}kl^2z}{4\left(l^4-z^4\right)}-\frac{i\sqrt{2}lm^2z^2}{\left(l^4-z^4\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}+\frac{\sqrt{2}kz^3}{4\left(l^4-z^4\right)}+\frac{\sqrt{2}l^2z}{l^4-z^4}+\frac{\sqrt{2}mz^2}{2\left(l^4-z^4\right)}
\end{aligned}
$$

$$\pi = -\frac{i\sqrt{2}\sqrt{l^2+z^2}k^2l^5}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-kl^2+kz^2+2\,mz}} + \frac{i\sqrt{2}\sqrt{l^2+z^2}k^2l^3z^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}k^2lz^4}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-kl^2+kz^2+2\,mz}}$$
$$+ \frac{\sqrt{2}kl^4z}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)} + \frac{i\sqrt{2}\sqrt{l^2+z^2}kl^3mz}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{i\sqrt{2}\sqrt{l^2+z^2}klmz^3}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{\sqrt{2}kz^5}{4\left(l^6+l^4z^2-l^2z^4-z^6\right)}$$
$$- \frac{\sqrt{2}l^4z}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}l^2mz^2}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)} - \frac{i\sqrt{2}\sqrt{l^2+z^2}lm^2z^2}{\left(l^6+l^4z^2-l^2z^4-z^6\right)\sqrt{-kl^2+kz^2+2\,mz}} - \frac{\sqrt{2}l^2z^3}{l^6+l^4z^2-l^2z^4-z^6} - \frac{\sqrt{2}mz^4}{2\left(l^6+l^4z^2-l^2z^4-z^6\right)}$$

$$\epsilon = 0$$

$$\gamma = -\frac{i\,k^2l^7vz}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{3i\,k^2l^5vz^3}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}$$
$$- \frac{3i\,k^2l^3vz^5}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{i\,k^2lvz^7}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}$$
$$- \frac{l^8H\frac{\partial}{\partial v}H}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)} - \frac{2\,l^6z^2H\frac{\partial}{\partial v}H}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{3\,l^4z^4H\frac{\partial}{\partial v}H}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{2\,l^2z^6H\frac{\partial}{\partial v}H}{l^8+2\,l^6z^2-2\,l^2z^6-z^8} - \frac{z^8H\frac{\partial}{\partial v}H}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)}$$
$$- \frac{i\,kl^7mv}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{3i\,kl^5mvz^2}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}$$
$$- \frac{7i\,kl^3mvz^4}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{5i\,klmvz^6}{2\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}$$
$$+ \frac{i\,l^5m^2vz}{\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}} + \frac{3i\,lm^2vz^5}{\left(l^8+2\,l^6z^2-2\,l^2z^6-z^8\right)\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$
\begin{aligned}
\Phi_{02} = &-\frac{k^4 l^{10}}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{k^4 l^8 z^2}{2\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&-\frac{3\,k^4 l^6 z^4}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{k^4 l^4 z^6}{2\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&-\frac{k^4 l^2 z^8}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{k^3 l^8 mz}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&-\frac{k^3 l^8 z^2}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}-\frac{3\,k^3 l^6 mz^3}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&+\frac{k^3 l^6 z^4}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{3\,k^3 l^4 mz^5}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&-\frac{k^3 l^2 mz^7}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}-\frac{k^3 l^2 z^8}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&+\frac{k^3 z^{10}}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}-\frac{k^2 l^{10}}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&+\frac{i\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}k^3 l^7 z}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{9\,k^2 l^8 z^2}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&-\frac{3\,k^2 l^6 m^2 z^2}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}-\frac{3i\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}k^3 l^5 z^3}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&+\frac{3\,k^2 l^6 mz^3}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}-\frac{2\,k^2 l^6 z^4}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&+\frac{6\,k^2 l^4 m^2 z^4}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}+\dots
\end{aligned}
$$

$$\Phi_{10} = 0$$

$$
\begin{aligned}
\Phi_{11} = & \frac{k^2 l^{14} \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{3\,k^2 l^{12} z^2 \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^{10} z^4 \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& - \frac{5\,k^2 l^8 z^6 \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& - \frac{5\,k^2 l^6 z^8 \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^4 z^{10} \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{3\,k^2 l^2 z^{12} \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 z^{14} \frac{\partial}{\partial v} H^2}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^{14} H \frac{\partial^2}{(\partial v)^2} H}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{3\,k^2 l^{12} z^2 H \frac{\partial^2}{(\partial v)^2} H}{4\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,kl^{12} mz + 12\,kl^8 mz^5 - 12\,kl^4 mz^9 + 4\,kmz^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{12} = & \frac{i\,\sqrt{2}\sqrt{-kl^2 + kz^2 + 2\,mz}\,k^2 l^{15} v \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})\sqrt{l^2 + z^2}} \\
& - \frac{\sqrt{2} k^2 l^{14} vz \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})} \\
& + \frac{3i\,\sqrt{2}\sqrt{-kl^2 + kz^2 + 2\,mz}\,k^2 l^{13} vz^2 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})\sqrt{l^2 + z^2}} \\
& - \frac{3\,\sqrt{2} k^2 l^{12} vz^3 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})} \\
& + \frac{i\,\sqrt{2}\sqrt{-kl^2 + kz^2 + 2\,mz}\,k^2 l^{11} vz^4 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})\sqrt{l^2 + z^2}} \\
& - \frac{\sqrt{2} k^2 l^{10} vz^5 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})} \\
& - \frac{5i\,\sqrt{2}\sqrt{-kl^2 + kz^2 + 2\,mz}\,k^2 l^9 vz^6 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})\sqrt{l^2 + z^2}} \\
& + \frac{5\,\sqrt{2} k^2 l^8 vz^7 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})} \\
& - \frac{5i\,\sqrt{2}\sqrt{-kl^2 + kz^2 + 2\,mz}\,k^2 l^7 vz^8 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})\sqrt{l^2 + z^2}} \\
& + \frac{5\,\sqrt{2} k^2 l^6 vz^9 \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})} \\
& + \frac{i\,\sqrt{2}\sqrt{-kl^2 + kz^2 + 2\,mz}\,k^2 l^5 vz^{10} \frac{\partial}{\partial v} H^2}{4\,(kl^{16} + 2\,kl^{14} z^2 - 2\,kl^{12} z^4 - 6\,kl^{10} z^6 + 6\,kl^6 z^{10} + 2\,kl^4 z^{12} - 2\,kl^2 z^{14} - kz^{16} - 2\,l^{14} mz - 6\,l^{12} mz^3 - 2\,l^{10} mz^5 + 10\,l^8 mz^7 + 10\,l^6 mz^9 - 2\,l^4 mz^{11} - 6\,l^2 mz^{13} - 2\,mz^{15})\sqrt{l^2 + z^2}} - \dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{20} = &-\frac{k^4 l^{10}}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{k^4l^8z^2}{2\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&-\frac{3\,k^4l^6z^4}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{k^4l^4z^6}{2\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&-\frac{k^4l^2z^8}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{k^3l^8mz}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&-\frac{k^3l^8z^2}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}-\frac{3\,k^3l^6mz^3}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&+\frac{k^3l^6z^4}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{3\,k^3l^4mz^5}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&-\frac{k^3l^2mz^7}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}-\frac{k^3l^2z^8}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&+\frac{k^3z^{10}}{8\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}-\frac{k^2l^{10}}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&-\frac{i\,\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}k^3l^7z}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}+\frac{9\,k^2l^8z^2}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&-\frac{3\,k^2l^6m^2z^2}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}+\frac{3i\,\sqrt{-kl^2+kz^2+2\,mz}\sqrt{l^2+z^2}k^3l^5z^3}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}\\
&+\frac{3\,k^2l^6mz^3}{4\,(kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11})}-\frac{2\,k^2l^6z^4}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}\\
&+\frac{6\,k^2l^4m^2z^4}{kl^{12}-3\,kl^8z^4+3\,kl^4z^8-kz^{12}-2\,l^{10}mz-2\,l^8mz^3+4\,l^6mz^5+4\,l^4mz^7-2\,l^2mz^9-2\,mz^{11}}-\dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{21} = &-\frac{i\,\sqrt{2}\sqrt{-kl^2+kz^2+2\,mz}k^2l^{15}v\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})\sqrt{l^2+z^2}}\\
&-\frac{\sqrt{2}k^2l^{14}vz\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})}\\
&-\frac{3i\,\sqrt{2}\sqrt{-kl^2+kz^2+2\,mz}k^2l^{13}vz^2\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})\sqrt{l^2+z^2}}\\
&-\frac{3\,\sqrt{2}k^2l^{12}vz^3\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})}\\
&-\frac{i\,\sqrt{2}\sqrt{-kl^2+kz^2+2\,mz}k^2l^{11}vz^4\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})\sqrt{l^2+z^2}}\\
&-\frac{\sqrt{2}k^2l^{10}vz^5\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})}\\
&+\frac{5i\,\sqrt{2}\sqrt{-kl^2+kz^2+2\,mz}k^2l^9vz^6\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})\sqrt{l^2+z^2}}\\
&+\frac{5\,\sqrt{2}k^2l^8vz^7\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})}\\
&+\frac{5i\,\sqrt{2}\sqrt{-kl^2+kz^2+2\,mz}k^2l^7vz^8\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})\sqrt{l^2+z^2}}\\
&+\frac{5\,\sqrt{2}k^2l^6vz^9\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})}\\
&-\frac{i\,\sqrt{2}\sqrt{-kl^2+kz^2+2\,mz}k^2l^5vz^{10}\,\frac{\partial}{\partial v}H^2}{4\,(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15})\sqrt{l^2+z^2}}-\dots
\end{aligned}
$$

$$
\begin{aligned}
\Phi_{22} = &-\frac{k^4 l^{18} v^2 \frac{\partial}{\partial v} H^2}{2\,(kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19})} \\
&+\frac{2\,k^4 l^{14} v^2 z^4 \frac{\partial}{\partial v} H^2}{kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19}} \\
&-\frac{3\,k^4 l^{10} v^2 z^8 \frac{\partial}{\partial v} H^2}{kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19}} \\
&+\frac{2\,k^4 l^6 v^2 z^{12} \frac{\partial}{\partial v} H^2}{kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19}} \\
&-\frac{k^4 l^2 v^2 z^{16} \frac{\partial}{\partial v} H^2}{2\,(kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19})} \\
&-\frac{k^4 l^{18} v^2 H \frac{\partial^2}{(\partial v)^2} H}{2\,(kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19})} \\
&+\frac{2\,k^4 l^{14} v^2 z^4 H \frac{\partial^2}{(\partial v)^2} H}{kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19}} \\
&-\frac{3\,k^4 l^{10} v^2 z^8 H \frac{\partial^2}{(\partial v)^2} H}{kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19}} \\
&+\frac{2\,k^4 l^6 v^2 z^{12} H \frac{\partial^2}{(\partial v)^2} H}{kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19}} \\
&-\frac{k^4 l^2 v^2 z^{16} H \frac{\partial^2}{(\partial v)^2} H}{2\,(kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19})} \\
&+\frac{4\,k^3 l^{16} m v^2 z \frac{\partial}{\partial v} H^2}{kl^{20} + 4\,kl^{18}z^2 + 3\,kl^{16}z^4 - 8\,kl^{14}z^6 - 14\,kl^{12}z^8 + 14\,kl^8z^{12} + 8\,kl^6z^{14} - 3\,kl^4z^{16} - 4\,kl^2z^{18} - kz^{20} - 2\,l^{18}mz - 10\,l^{16}mz^3 - 16\,l^{14}mz^5 + 28\,l^{10}mz^9 + 28\,l^8mz^{11} - 16\,l^4mz^{15} - 10\,l^2mz^{17} - 2\,mz^{19}} + \cdots
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = &-\frac{k^2 l^{14} \frac{\partial}{\partial v} H^2}{12\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&-\frac{k^2 l^{12} z^2 \frac{\partial}{\partial v} H^2}{4\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&-\frac{k^2 l^{10} z^4 \frac{\partial}{\partial v} H^2}{12\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&+\frac{5\,k^2 l^8 z^6 \frac{\partial}{\partial v} H^2}{12\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&+\frac{5\,k^2 l^6 z^8 \frac{\partial}{\partial v} H^2}{12\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&-\frac{k^2 l^4 z^{10} \frac{\partial}{\partial v} H^2}{12\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&-\frac{k^2 l^2 z^{12} \frac{\partial}{\partial v} H^2}{4\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&-\frac{k^2 z^{14} \frac{\partial}{\partial v} H^2}{12\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&-\frac{k^2 l^{14} H \frac{\partial^2}{(\partial v)^2} H}{12\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} \\
&-\frac{k^2 l^{12} z^2 H \frac{\partial^2}{(\partial v)^2} H}{4\,(k^2l^{14} - k^2l^{12}z^2 - 3\,k^2l^{10}z^4 + 3\,k^2l^8z^6 + 3\,k^2l^6z^8 - 3\,k^2l^4z^{10} - k^2l^2z^{12} + k^2z^{14} - 4\,kl^{12}mz + 12\,kl^8mz^5 - 12\,kl^4mz^9 + 4\,kmz^{13} + 4\,l^{10}m^2z^2 + 4\,l^8m^2z^4 - 8\,l^6m^2z^6 - 8\,l^4m^2z^8 + 4\,l^2m^2z^{10} + 4\,m^2z^{12})} - \cdots
\end{aligned}
$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = {} & \frac{k^2 l^{14} \frac{\partial}{\partial v} H^2}{6\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^{12} z^2 \frac{\partial}{\partial v} H^2}{2\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^{10} z^4 \frac{\partial}{\partial v} H^2}{6\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& - \frac{5\,k^2 l^8 z^6 \frac{\partial}{\partial v} H^2}{6\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& - \frac{5\,k^2 l^6 z^8 \frac{\partial}{\partial v} H^2}{6\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^4 z^{10} \frac{\partial}{\partial v} H^2}{6\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^2 z^{12} \frac{\partial}{\partial v} H^2}{2\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 z^{14} \frac{\partial}{\partial v} H^2}{6\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^{14} H \frac{\partial^2}{(\partial v)^2} H}{6\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \frac{k^2 l^{12} z^2 H \frac{\partial^2}{(\partial v)^2} H}{2\,(k^2 l^{14} - k^2 l^{12} z^2 - 3\,k^2 l^{10} z^4 + 3\,k^2 l^8 z^6 + 3\,k^2 l^6 z^8 - 3\,k^2 l^4 z^{10} - k^2 l^2 z^{12} + k^2 z^{14} - 4\,k l^{12} m z + 12\,k l^8 m z^5 - 12\,k l^4 m z^9 + 4\,k m z^{13} + 4\,l^{10} m^2 z^2 + 4\,l^8 m^2 z^4 - 8\,l^6 m^2 z^6 - 8\,l^4 m^2 z^8 + 4\,l^2 m^2 z^{10} + 4\,m^2 z^{12})} \\
& + \dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_3 = &-\frac{\sqrt{2}k^2l^{14}vz\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{3\,\sqrt{2}k^2l^{12}vz^3\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{\sqrt{2}k^2l^{10}vz^5\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&+\frac{5\,\sqrt{2}k^2l^8vz^7\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&+\frac{5\,\sqrt{2}k^2l^6vz^9\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{\sqrt{2}k^2l^4vz^{11}\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{3\,\sqrt{2}k^2l^2vz^{13}\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{\sqrt{2}k^2vz^{15}\frac{\partial}{\partial v}H^2}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{\sqrt{2}k^2l^{14}vzH\frac{\partial^2}{(\partial v)^2}H}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{3\,\sqrt{2}k^2l^{12}vz^3H\frac{\partial^2}{(\partial v)^2}H}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&-\frac{\sqrt{2}k^2l^{10}vz^5H\frac{\partial^2}{(\partial v)^2}H}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&+\frac{5\,\sqrt{2}k^2l^8vz^7H\frac{\partial^2}{(\partial v)^2}H}{4\left(kl^{16}+2\,kl^{14}z^2-2\,kl^{12}z^4-6\,kl^{10}z^6+6\,kl^6z^{10}+2\,kl^4z^{12}-2\,kl^2z^{14}-kz^{16}-2\,l^{14}mz-6\,l^{12}mz^3-2\,l^{10}mz^5+10\,l^8mz^7+10\,l^6mz^9-2\,l^4mz^{11}-6\,l^2mz^{13}-2\,mz^{15}\right)}\\
&+\dots
\end{aligned}
$$

$$
\begin{aligned}
\Psi_4 = {} & \frac{k^4 l^{10} v^2 \frac{\partial}{\partial v} H^2}{2\left(kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}\right)} \\
& - \frac{2\,k^4 l^8 v^2 z^2 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& + \frac{3\,k^4 l^6 v^2 z^4 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& - \frac{2\,k^4 l^4 v^2 z^6 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& + \frac{k^4 l^2 v^2 z^8 \frac{\partial}{\partial v} H^2}{2\left(kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}\right)} \\
& + \frac{k^4 l^{10} v^2 H \frac{\partial^2}{(\partial v)^2} H}{2\left(kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}\right)} \\
& - \frac{2\,k^4 l^8 v^2 z^2 H \frac{\partial^2}{(\partial v)^2} H}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& + \frac{3\,k^4 l^6 v^2 z^4 H \frac{\partial^2}{(\partial v)^2} H}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& - \frac{2\,k^4 l^4 v^2 z^6 H \frac{\partial^2}{(\partial v)^2} H}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& + \frac{k^4 l^2 v^2 z^8 H \frac{\partial^2}{(\partial v)^2} H}{2\left(kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}\right)} \\
& - \frac{4\,k^3 l^8 m v^2 z \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& + \frac{k^3 l^8 v^2 z^2 \frac{\partial}{\partial v} H^2}{2\left(kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}\right)} \\
& + \frac{12\,k^3 l^6 m v^2 z^3 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& - \frac{k^3 l^6 v^2 z^4 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& - \frac{12\,k^3 l^4 m v^2 z^5 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& + \frac{4\,k^3 l^2 m v^2 z^7 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& + \frac{k^3 l^2 v^2 z^8 \frac{\partial}{\partial v} H^2}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} \\
& - \frac{k^3 v^2 z^{10} \frac{\partial}{\partial v} H^2}{2\left(kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}\right)} \\
& - \frac{4\,k^3 l^8 m v^2 z H \frac{\partial^2}{(\partial v)^2} H}{kl^{12} - 3\,kl^8 z^4 + 3\,kl^4 z^8 - kz^{12} - 2\,l^{10} mz - 2\,l^8 mz^3 + 4\,l^6 mz^5 + 4\,l^4 mz^7 - 2\,l^2 mz^9 - 2\,mz^{11}} + \cdots
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"